that Veneziano and others had been poring over would be mathematically described using Euler's formula. The little elastic strands were christened *strings* and now, with the horse properly before the cart, string theory was officially born.

But hold the bubbly. To those involved in this research, it was gratifying to understand the physical origin of Veneziano's insight, since it suggested that physicists were on their way to unmasking the strong nuclear force. Yet the discovery was not greeted with universal enthusiasm; far from it. Very far. In fact, Susskind's paper was returned by the journal to which he submitted it with the comment that the work was of minimal interest, an evaluation Susskind recalls well: "I was stunned, I was knocked off my chair, I was depressed, so I went home and got drunk."[9] Eventually, his paper and the others that announced the string concept were all published, but it was not long before the theory suffered two more devastating setbacks. Close scrutiny of more refined data on the strong nuclear force, collected during the early 1970s, showed that the string approach failed to describe the newer results accurately. Moreover, a new proposal called *quantum chromodynamics*, which was firmly rooted in the traditional ingredients of particles and fields—no strings at all—*was* able to describe all the data convincingly. And so by 1974, string theory had been dealt a one-two knockout punch. Or so it seemed.

John Schwarz was one of the earliest string enthusiasts. He once told me that from the start, he had a gut feeling that the theory was deep and important. Schwarz spent a number of years analyzing its various mathematical aspects; among other things, this led to the discovery of *super*string theory—as we shall see, an important refinement of the original string proposal. But with the rise of quantum chromodynamics and the failure of the string framework to describe the strong force, the justification for continuing to work on string theory began to run thin. Nevertheless, there was one particular mismatch between string theory and the strong nuclear force that kept nagging at Schwarz, and he found that he just couldn't let it go. The quantum mechanical equations of string theory predicted that a particular, rather unusual, particle should be copiously produced in the high-energy collisions taking place in atom smashers. The particle would have zero mass, like a photon, but string theory predicted it would have *spin-two*, meaning, roughly speaking, that